Mock Exam 3 — Version A — Solutions

1. Curl, divergence, and conservativity

Key observation. The $\sin z$ and $\cos z$ terms come from the gradient of $xy\sin z$; the remaining planar rotation supplies a nonzero curl.

Writing $\mathbf{F} = (P, Q, R)$ gives

$$\begin{aligned} R_y - Q_z &= x\cos z - x\cos z = 0, \\ P_z - R_x &= y\cos z - y\cos z = 0, \\ Q_x - P_y &= (\sin z + 1) - (\sin z - 1) = 2. \end{aligned}$$

Therefore

$$\boxed{\nabla \times \mathbf{F} = (0, 0, 2)}, \qquad \boxed{\nabla \cdot \mathbf{F} = -xy\sin z}.$$

The divergence follows from $P_x = Q_y = 0$ and $R_z = -xy\sin z$. The field is **not conservative**: a gradient field with continuous second partial derivatives has zero curl, whereas this curl is nonzero everywhere.

Check. The decomposition $\mathbf{F} = \nabla(xy\sin z) + (-y, x, 0)$ independently gives the curl $(0, 0, 2)$ and divergence $-xy\sin z$.

Alternative. Computing all six mixed-component derivatives directly is fully sufficient; recognizing the decomposition is optional.

2. Green's theorem

Key observation. The exponential terms have equal mixed derivatives. After they cancel, the integral depends only on areas and a first moment of an off-center hole. The outer disk's odd moment is zero, but the hole's moment is not.

Let $P = ye^{xy} - y$ and $Q = xe^{xy} + 3x^2 + x$. Then

$$Q_x = e^{xy} + xye^{xy} + 6x + 1, \qquad P_y = e^{xy} + xye^{xy} - 1,$$

so $Q_x - P_y = 6x + 2$. The given orientation is positive: the outer circle is counterclockwise and the inner circle clockwise. Green's theorem gives

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_D (6x + 2)\,\mathrm{d}A.$$

The smaller disk lies strictly inside the larger one because $1 + 1/2 < 2$. An explicit setup is

$$\begin{aligned} \iint_D (6x + 2)\,\mathrm{d}A = &\int_0^{2\pi}\int_0^{2} (6r\cos\theta + 2) r\,\mathrm{d}r\,\mathrm{d}\theta \\ &- \int_0^{2\pi}\int_0^{1/2} \big(6(1 + s\cos\theta) + 2\big) s\,\mathrm{d}s\,\mathrm{d}\theta. \end{aligned}$$

The second integral uses coordinates centered at $(1, 0)$, not at the origin. Terms containing $\cos\theta$ integrate to zero. Thus

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = 2(4\pi) - 8\left(\frac{\pi}{4}\right) = \boxed{6\pi}.$$

Check. The area of D is $15\pi/4$ and $\iint_D x \, \mathrm{d}A = 0-(1)(\pi/4) = -\pi/4$; hence $6(-\pi/4)+2(15\pi/4) = 6\pi$.

Alternative. Cartesian bounds, splitting the vertical slices through the hole, are valid. Translating the small disk and using its symmetry avoids that split. A proof of the first-moment calculation by integrating in the displayed translated coordinates is already given above.

3. Three surface parametrizations

Key observation. Use coordinates suited to the geometry, then translate every extra condition into a parameter bound. No surface-area calculation is needed.

(a) In polar coordinates the cylinder condition becomes

$$(r\cos\theta - 1)^2 + r^2\sin^2\theta \le 1 \quad \Longleftrightarrow \quad r(r - 2\cos\theta) \le 0.$$

Thus one parametrization is

$$\boxed{\mathbf{r}(r,\theta) = (r\cos\theta, r\sin\theta, r)}, \qquad \begin{cases} -\pi/2 \le \theta \le \pi/2, \\ 0 \le r \le 2\cos\theta. \end{cases}$$

The cone's tip is a boundary point; degeneracy of polar coordinates there does not affect the image.

(b) Scale spherical coordinates to the ellipsoid:

$$\boxed{\mathbf{r}(\phi,\theta) = (2\sin\phi\cos\theta, 3\sin\phi\sin\theta, \cos\phi)}, \qquad \begin{cases} 0 \le \phi \le \pi/4, \\ 0 \le \theta \le \pi. \end{cases}$$

The cone-like inequality is $\cos\phi \ge \sin\phi$, so it forces $0 \le \phi \le \pi/4$; $y \ge 0$ then gives the stated azimuthal range.

(c) A parametrization using the taught y-axis revolution convention is

$$\boxed{\mathbf{r}(s,\theta) = \big((2+\cos s)\cos\theta, \sin s, -(2+\cos s)\sin\theta\big)},$$

$$\begin{cases} -\pi/2 \le s \le \pi/2, \\ 0 \le \theta \le 2\pi. \end{cases}$$

Here the distance to the y-axis is $2 + \cos s \ge 2$, so this is an ordinary surface of revolution without a crossing of the axis.

Check. In (a), $z = r = \sqrt{x^2+y^2}$. In (b), substitution gives $x^2/4 + y^2/9 + z^2 = 1$ exactly. In (c), setting $\theta = 0$ recovers the given generating arc, and varying θ preserves its height and distance from the axis.

Alternative. Any two-parameter descriptions with the same images and correct complete bounds are valid. Reversing angular direction, shifting angles by 2π, using a different pair of parameter names, and different once-around representations are equivalent. No orientation is requested.

4. A scalar surface integral: setup only

Key observation. On the supplied surface, $x^2 - y^2 = 4z$, so the exponential collapses to 1 before the scalar integrand is multiplied by the area factor.

The given parametrization has

$$\mathbf{r}_u = (1, 1, v), \qquad \mathbf{r}_v = (1, -1, u),$$

$$\mathbf{r}_u \times \mathbf{r}_v = (u + v, v - u, -2), \qquad \|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{4 + 2u^2 + 2v^2}.$$

The scalar substitution is

$$f(\mathbf{r}(u, v)) = e^{(u+v)^2-(u-v)^2-4uv} + uv = 1 + uv.$$

Consequently the requested setup is

$$\boxed{\iint_S f \, \mathrm{d}\sigma = \int_0^1 \int_0^{1-u} (1 + uv)\sqrt{4 + 2u^2 + 2v^2} \, \mathrm{d}v \, \mathrm{d}u}.$$

Check. The squared cross-product norm is $(u + v)^2 + (v - u)^2 + 4 = 4 + 2u^2 + 2v^2 > 0$. The first two surface coordinates uniquely recover $u = (x + y)/2$ and $v = (x - y)/2$, so the patch is not multiply covered. The density $1 + uv$ is nonnegative on the parameter triangle.

Alternative. Reversing the integration order gives bounds $0 \le v \le 1$, $0 \le u \le 1 - v$ with the same scalar integrand. The opposite cross-product order is valid because its magnitude is unchanged.

5. Direct flux through a curved surface

Key observation. The terms involving e^{z^2} describe motion around the axis, so their normal contribution cancels. A second cancellation uses $z = x^2+y^2$; the surviving integral has the derivative needed for e^{r^2}.

Choose

$$\mathbf{r}(r, \theta) = (r\cos\theta, r\sin\theta, r^2), \qquad \begin{cases} 1 \le r \le 2, \\ 0 \le \theta \le 2\pi. \end{cases}$$

The tangent vectors and outward cross product are

$$\mathbf{r}_r = (\cos\theta, \sin\theta, 2r), \qquad \mathbf{r}_\theta = (-r\sin\theta, r\cos\theta, 0),$$

$$\mathbf{n} = \mathbf{r}_\theta \times \mathbf{r}_r = (2r^2\cos\theta, 2r^2\sin\theta, -r).$$

At $(r, \theta) = (1, 0)$ this gives $(2, 0, -1)$, whose radial component points away from the z-axis. More generally, $\mathbf{n} \cdot (r\cos\theta, r\sin\theta, 0) = 2r^3 > 0$, verifying the required orientation.

Set $c = \cos\theta$, $s = \sin\theta$. Substitution gives

$$\mathbf{F}(\mathbf{r}) = \left(rc - rse^{r^4},\ rs + rce^{r^4},\ 2r^2 - e^{r^2}\right).$$

The dot product simplifies before any integration:

$$\begin{aligned} \mathbf{F}(\mathbf{r}) \cdot \mathbf{n} &= 2r^3c^2 - 2r^3sce^{r^4} + 2r^3s^2 + 2r^3sce^{r^4} - 2r^3 + re^{r^2} \\ &= re^{r^2}. \end{aligned}$$

Thus

$$\boxed{\iint_S \mathbf{F}\cdot\mathbf{N}\,\mathrm{d}\sigma = \int_0^{2\pi}\int_1^2 re^{r^2}\,\mathrm{d}r\,\mathrm{d}\theta = \pi(e^4-e)}.$$

Check. The vector $(x, y, 2z)$ is tangent to $z = x^2 + y^2$ because its dot product with $(2x, 2y, -1)$ is zero on that surface. The rotation vector $(-y, x, 0)$ is also tangent. Only $(0, 0, -e^z)$ contributes to the outward flux, and its downward component agrees with the downward component of the outward normal, yielding positive flux.

Alternative. A direct graph parametrization $(x, y, x^2 + y^2)$ with outward vector $(2x, 2y, -1)$ produces integrand $e^{x^2+y^2}$ over $1 \le x^2 + y^2 \le 4$. Converting that double integral to polar gives the same result. Both methods directly parametrize the stated surface.

6. Stokes' theorem and the whole boundary

Key observation. The lateral cone has two boundary circles, at different heights. Their induced directions are opposite, and the coefficient $z+1$ takes a different value on each. The third component of the field contributes nothing because both circles have $\mathrm{d}z = 0$.

An outward normal on the upper cone is proportional to $(\cos\theta, \sin\theta, -1)$, so it points downward. Consequently the outer circle at $z = 3$ is clockwise from above, and the inner circle at $z = 1$ is counterclockwise.

One-parameter parametrizations with the induced orientations are

$$\begin{aligned}\mathbf{c}_o(t) &= (3\cos t, -3\sin t, 3),\\ \mathbf{c}_i(t) &= (\cos t, \sin t, 1),\end{aligned}\qquad 0 \le t \le 2\pi.$$

Their derivatives and substituted fields are

$$\begin{aligned}\mathbf{c}_o'(t) &= (-3\sin t, -3\cos t, 0), & \mathbf{F}(\mathbf{c}_o(t)) &= (12\sin t, 12\cos t, e^{27}),\\ \mathbf{c}_i'(t) &= (-\sin t, \cos t, 0), & \mathbf{F}(\mathbf{c}_i(t)) &= (-2\sin t, 2\cos t, e).\end{aligned}$$

Thus

$$\mathbf{F}(\mathbf{c}_o(t))\cdot\mathbf{c}_o'(t) = -36, \qquad \mathbf{F}(\mathbf{c}_i(t))\cdot\mathbf{c}_i'(t) = 2.$$

Stokes' theorem, with the whole oriented boundary, gives

$$\begin{aligned}\iint_S (\nabla\times\mathbf{F})\cdot\mathbf{N}\,\mathrm{d}\sigma &= \oint_{C_o}\mathbf{F}\cdot\mathrm{d}\mathbf{r} + \oint_{C_i}\mathbf{F}\cdot\mathrm{d}\mathbf{r}\\ &= \int_0^{2\pi}(-36)\,\mathrm{d}t + \int_0^{2\pi} 2\,\mathrm{d}t = \boxed{-68\pi}.\end{aligned}$$

Check. On a counterclockwise circle of radius R at height h, the first two field components give circulation $2\pi(h+1)R^2$. Reversing the outer circle therefore gives $-2\pi(4)(9) + 2\pi(2)(1) = -68\pi$, confirming both signs and radii.

Alternative. Using counterclockwise parametrizations for both circles and subtracting the outer-circle integral is equivalent. A direct surface calculation also verifies the number independently.

7. Conceptual decisions

(a)

Expression	Type	Necessarily zero?
$\nabla \times (f\mathbf{G})$	vector field	No
$\nabla \cdot (\nabla \times \mathbf{G})$	scalar field	Yes
$\nabla \times (\nabla \cdot \mathbf{G})$	undefined	not applicable

For the first row, $f = 1$ and $\mathbf{G} = (-y, x, 0)$ give curl $(0, 0, 2)$, so a universal zero claim is false. The second row is the identity $\nabla \cdot (\nabla \times \mathbf{G}) = 0$. In the third row, divergence produces a scalar, but curl takes a vector field as its input.

(b) The only directly applicable listed method is **direct parametrization**. The curve is open, so Green's and Stokes' theorems do not directly apply. Although

$$\nabla \times (0, xy, 0) = (0, 0, y)$$

vanishes along C, it is nonzero at points with $y \neq 0$, including points arbitrarily close to the curve. Hence the stated field is not conservative on an open neighborhood of C, and FTLI cannot be invoked for it. The fact that its particular line integral happens to equal zero does not supply the missing FTLI hypothesis.

(c) The negative divergence indicates a local **sink**, or local net inflow. The directed rotation axis is **positive** z, and the right-hand rule gives **counterclockwise** rotation viewed from the positive z-axis. These are local conclusions at p, not global statements about the flow.

Check. For (b), direct substitution gives $\mathbf{F}(t, 0, 0) = \mathbf{0}$, so direct evaluation is possible even though FTLI is not applicable to the given nonconservative field. For (c), the signs and axis are read from the supplied values, without needing a field formula.

Alternative. Any equivalent physical wording for net local inflow and the positive-z right-hand-rule sense is acceptable. No global behavior or trajectory shape is determined by those two local measurements.

Mock Exam 3 — Version B — Solutions

1. Curl, divergence, and conservativity

Key observation. The exponential and trigonometric pieces have matching mixed partials. Their cancellations must be performed before deciding whether the remaining curl vanishes everywhere.

Writing $\mathbf{F} = (P, Q, R)$ gives

$$\begin{aligned} R_y - Q_z &= 2z - (2z + x) = -x, \\ P_z - R_x &= (\cos x - y) - \cos x = -y, \\ Q_x - P_y &= (e^{xy} + xye^{xy} + z) - (e^{xy} + xye^{xy} - z) = 2z. \end{aligned}$$

Consequently

$$\boxed{\nabla \times \mathbf{F} = (-x, -y, 2z)}, \qquad \boxed{\nabla \cdot \mathbf{F} = (x^2 + y^2)e^{xy} - z\sin x + 2y}.$$

The field is **not conservative** on $\mathbb{R}^3$, since its curl is not identically zero; for example, it equals $(-1, 0, 0)$ at $(1, 0, 0)$. Its zero curl at the origin alone cannot establish conservativity.

Check. The first three terms form $\nabla(e^{xy} + z\sin x + yz^2)$. The remaining field $(-yz, xz, 0)$ has curl $(-x, -y, 2z)$ and divergence zero, independently confirming both the cancellation structure and the final formulas.

Alternative. Differentiate the three components directly, as above. A potential need not be found because the nonzero curl already rules one out.

2. Green's theorem on an unfinished curve

Key observation. Close the missing diagonal, use the clockwise orientation of the resulting curve, and subtract the nonzero integral on that diagonal.

Let L be the segment from $(2, 1)$ back to $(1, 0)$. The closed curve $C+L$ traverses its region clockwise. The enclosed region is

$$D: \qquad \begin{cases} 0 \le y \le 1, \\ y^2 \le x \le 1 + y. \end{cases}$$

For $P = ye^{xy} - y^2/2$ and $Q = xe^{xy} + 2x$,

$$Q_x - P_y = (e^{xy} + xye^{xy} + 2) - (e^{xy} + xye^{xy} - y) = 2 + y.$$

Green's theorem, with the clockwise sign, gives

$$\begin{aligned} \int_C \mathbf{F}\cdot d\mathbf{r} + \int_L \mathbf{F}\cdot d\mathbf{r} &= -\int_0^1 \int_{y^2}^{1+y} (2+y)\, dx\, dy \\ &= -\int_0^1 (2 + 3y - y^2 - y^3)\, dy = -\frac{35}{12}. \end{aligned}$$

Parametrize the closing segment by $\ell(t) = (1 + t, t)$, with t decreasing from 1 to 0. Then

$$\begin{aligned} \int_L \mathbf{F}\cdot d\mathbf{r} &= \int_1^0 \left((1+2t)e^{t+t^2} - \frac{t^2}{2} + 2 + 2t \right) dt \\ &= \left[e^{t+t^2} - \frac{t^3}{6} + 2t + t^2 \right]_1^0 = -e^2 - \frac{11}{6}. \end{aligned}$$

Therefore

$$\boxed{\int_C \mathbf{F}\cdot d\mathbf{r} = -\frac{35}{12} - \left(-e^2 - \frac{11}{6}\right) = e^2 - \frac{13}{12}}.$$

Check. For an independent check, the original three pieces have integrals 0, $e-7/12$, and $e^2-e-1/2$. Their sum agrees with the answer. This directly confirms the result.

Alternative. One may close by an easier two-segment polygonal route, provided every auxiliary integral and the resulting oriented region are accounted for. The diagonal gives a single closing integral.

3. Surface parametrizations and intersection bounds

Key observation. The equations determine parameter bounds by intersections, not just by identifying a familiar surface.

(a) One parametrization is

$$\boxed{\mathbf{r}(\theta,\phi) = (2\sin\phi\cos\theta, 2\sin\phi\sin\theta, 2\cos\phi)}, \qquad \boxed{0 \le \theta \le 2\pi, \quad \pi/6 \le \phi \le \pi/2}.$$

The outside-cylinder condition is $4\sin^2\phi \ge 1$. Since $z \ge 0$, $0 \le \phi \le \pi/2$, so this becomes $\phi \ge \pi/6$.

(b) A convenient parametrization and bounds are

$$\boxed{\mathbf{r}(\theta, v) = (\cos\theta, \sin\theta, v)}, \qquad \boxed{0 \le \theta \le 2\pi, \quad \sin\theta \le v \le 2 + \cos\theta}.$$

The upper plane is always above the lower one because $2 + \cos\theta - \sin\theta \ge 2 - \sqrt{2} > 0$.

(c) Using the radius as a parameter,

$$\boxed{\mathbf{r}(u,\theta) = (u\cos\theta, u\sin\theta, u)}, \qquad \boxed{1 \le u \le \sqrt{2}, \quad 0 \le \theta \le 2\pi}.$$

On the cone, the sphere condition becomes $2u^2 \le 4$, while $z \ge 1$ becomes $u \ge 1$.

Check. Substitution gives the sphere, cylinder, and cone equations respectively. Their bounds include each required limiting intersection and add no caps.

Alternative. In (b), one can instead use

$$\mathbf{r}(\theta, t) = (\cos\theta, \sin\theta, \sin\theta + t(2 + \cos\theta - \sin\theta)),$$

with $0 \le \theta \le 2\pi$, $0 \le t \le 1$. Other correct two-parameter maps with equivalent domains are valid.

4. Set up a scalar surface integral

Key observation. The given coordinates make $y - x^2 = v$ and $z/x = v$. The latter identity is legitimate because $x = u \ge 1$ everywhere on the patch.

The tangent vectors and cross product are

$$\begin{aligned} \mathbf{r}_u &= (1, 2u, v), \qquad \mathbf{r}_v = (0, 1, u), \\ \mathbf{r}_u \times \mathbf{r}_v &= (2u^2 - v, -u, 1), \\ |\mathbf{r}_u \times \mathbf{r}_v| &= \sqrt{(2u^2 - v)^2 + u^2 + 1}. \end{aligned}$$

Substitution gives

$$f(\mathbf{r}(u,v)) = e^{(u^2+v)-u^2} + \left(\frac{uv}{u}\right)^2 = e^v + v^2.$$

Thus the requested setup is

$$\boxed{\int_1^2 \int_0^{2-u} (e^v + v^2)\sqrt{(2u^2 - v)^2 + u^2 + 1}\ \mathrm{d}v\ \mathrm{d}u}.$$

It should not be evaluated.

Check. The third component of the cross product is 1, so the patch is regular throughout its domain. Also $u = x$ and $v = y - x^2$ recover the parameters uniquely; the map does not cover the surface multiple times.

Alternative. Reversing the cross-product order gives the same magnitude. The equivalent bounds $0 \le v \le 1$, $1 \le u \le 2 - v$ are also valid.

5. Flux through a curved sheet

Key observation. The first-quadrant lens is unchanged by exchanging x and y. After forming the correctly oriented flux integrand, its antisymmetric part integrates to zero.

The two boundary curves meet at $(0,0)$ and $(1,1)$. The region has bounds

$$D: \qquad \begin{cases} 0 \le x \le 1, \\ x^2 \le y \le \sqrt{x}. \end{cases}$$

Use $\mathbf{r}(x,y) = (x, y, xy)$. Then

$$\mathbf{r}_x = (1, 0, y), \qquad \mathbf{r}_y = (0, 1, x), \qquad \mathbf{r}_x \times \mathbf{r}_y = (-y, -x, 1).$$

This cross product points upward. The required downward oriented area vector is therefore $\mathbf{r}_y \times \mathbf{r}_x = (y, x, -1)$; its third component is negative at every point.

The field and dot product become

$$\begin{aligned} \mathbf{F}(\mathbf{r}) &= (x + 1, y - 1, 3xy + x - y), \\ \mathbf{F}(\mathbf{r}) \cdot (y, x, -1) &= (x+1)y + (y-1)x - (3xy + x - y) \\ &= -xy - 2x + 2y. \end{aligned}$$

The complete flux setup is

$$\iint_S \mathbf{F} \cdot \mathbf{N}\ \mathrm{d}\sigma = \int_0^1 \int_{x^2}^{\sqrt{x}} (-xy - 2x + 2y)\ \mathrm{d}y\ \mathrm{d}x.$$

Because D is symmetric under $x \leftrightarrow y$, $\iint_D (-2x + 2y)\ \mathrm{d}A = 0$. Hence

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{N}\ \mathrm{d}\sigma &= -\int_0^1 \int_{x^2}^{\sqrt{x}} xy\ \mathrm{d}y\ \mathrm{d}x \\ &= -\frac{1}{2}\int_0^1 (x^2 - x^5)\ \mathrm{d}x = -\frac{1}{2}\left(\frac{1}{3} - \frac{1}{6}\right) = \boxed{-\frac{1}{12}}. \end{aligned}$$

Check. Direct integration gives $\iint_D x\ \mathrm{d}A = \iint_D y\ \mathrm{d}A = 3/20$ and $\iint_D xy\ \mathrm{d}A = 1/12$, confirming the cancellation and sign.

Alternative. Evaluate the displayed double integral without symmetry, or reverse the order using $0 \le y \le 1$, $y^2 \le x \le \sqrt{y}$. Both retain the direct surface-parametrization method.

6. Stokes' theorem with a nonplanar boundary

Key observation. On the boundary $x^2 + y^2 = 1$, two complicated field terms vanish. The remaining third component becomes a function of z alone, so its line-integral contribution is a total derivative around a closed curve.

The boundary is

$$C: \qquad x^2 + y^2 = 1, \qquad z = x^2 - y^2.$$

The upward orientation induces counterclockwise traversal of its xy-projection. Thus a matching one-parameter parametrization is

$$\mathbf{c}(t) = (\cos t, \sin t, \cos 2t), \qquad 0 \le t \le 2\pi,$$

with

$$\mathbf{c}\,'(t) = (-\sin t, \cos t, -2\sin 2t).$$

Stokes' theorem gives

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N}\, \mathrm{d}\sigma = \int_0^{2\pi} \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}\,'(t)\, \mathrm{d}t.$$

Put $z(t) = \cos 2t$. Since $1 - x^2 - y^2 = 0$ on C,

$$\mathbf{F}(\mathbf{c}(t)) = \big(-\sin t\, z(t)^2,\ \cos t\, z(t)^2,\ e + z(t)e^{z(t)}\big).$$

Consequently the dot product is

$$z(t)^2 + (e + z(t)e^{z(t)})z'(t) = \cos^2 2t - 2\sin 2t\, \left(e + \cos 2t\, e^{\cos 2t}\right).$$

The primitive

$$\Phi(z) = ez + (z-1)e^z \quad \text{satisfies} \quad \Phi'(z) = e + ze^z.$$

Since $z(0) = z(2\pi) = 1$, the second term integrates to zero. Therefore

$$\boxed{\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N}\, \mathrm{d}\sigma = \int_0^{2\pi} \cos^2 2t\ \mathrm{d}t + [\Phi(z(t))]_0^{2\pi} = \pi}.$$

Check. The terms multiplied by $1 - x^2 - y^2$ are exactly zero on every boundary point. For the remaining field, a separate surface check has integrand $6(x^2 - y^2)^2 - 8xye^{x^2+y^2}$ over the unit disk. The odd term vanishes and the first term integrates to π. This confirms the answer.

Alternative. Split the boundary integral into $\int z^2(-y\ \mathrm{d}x + x\ \mathrm{d}y)$ and $\int (e + ze^z)\ \mathrm{d}z$. The second is an exact one-variable differential, and the first becomes $\int_0^{2\pi} \cos^2 2t\ \mathrm{d}t$. This is the same Stokes-to-boundary method in differential notation.

7. Operators, physical meaning, and choice of method

(a) In order, the answers are

$$\boxed{\text{vector}, \quad \text{scalar}, \quad \text{undefined}, \quad \text{scalar}}.$$

Both ∇f and $\mathbf{F}$ are vectors, so their cross product is a vector and its curl is a vector. A dot product of two vectors is scalar. The divergence of $\mathbf{F}$ is scalar, so taking its curl is undefined. The curl of $\mathbf{F}$ is vector-valued, and taking its divergence is well-defined and scalar (in fact, identically zero).

(b) For $\mathbf{V} = (-y, x, z)$,

$$\nabla \cdot \mathbf{V} = 1 > 0, \qquad \nabla \times \mathbf{V} = (0, 0, 2).$$

Thus there is **local volume expansion**, and the local rotation is **counterclockwise viewed from the positive z-axis**. The field vanishes at the origin, but these derivative-based local behaviors do not.

(c) All four methods apply. The curve is smooth, closed, and lies in $z = 0$, while the field is smooth on all of $\mathbb{R}^3$. Direct parametrization therefore applies. Also

$$\mathbf{W} = \nabla \left(x^2 y + \frac{z^2}{2} \right),$$

so FTLI applies. On $z = 0$, the first two components form the smooth planar field $(2xy, x^2)$, so Green's theorem applies over the unit disk. The same upward-oriented disk is a spanning surface for Stokes' theorem. Both the planar scalar curl and the full curl vanish. No integral value is requested; every method would give zero.

Check. The potential's three derivatives are $(2xy, x^2, z)$. For the physical part, the positive z-component of curl follows the right-hand rule for the planar pair $(-y, x)$.

Alternative. For (c), checking zero curl on all of $\mathbb{R}^3$ also establishes conservativity, since this domain is simply connected; the explicit potential is the shortest justification.

Mock Exam 3 — Version C — Solutions

1. Curl, divergence, and conservativity

Key observation. The three rational terms have the same positive denominator. Their mixed derivatives cancel, and the polynomial part is also a gradient.

Put $q = x^2 + y^2 + z^2$, so $D = 1 + q$. If $\mathbf{F} = (P, Q, R)$, then

$$P_y = Q_x = z - \frac{4xy}{D^2}, \qquad P_z = R_x = y - \frac{4xz}{D^2}, \qquad Q_z = R_y = x - \frac{4yz}{D^2}.$$

Consequently

$$\boxed{\nabla \times \mathbf{F} = \mathbf{0}}.$$

For the divergence,

$$\begin{aligned} \nabla \cdot \mathbf{F} &= \left(\frac{2}{D} - \frac{4x^2}{D^2}\right) + \left(\frac{2}{D} - \frac{4y^2}{D^2}\right) + \left(\frac{2}{D} - \frac{4z^2}{D^2}\right) \\ &= \frac{6}{D} - \frac{4q}{D^2} = \boxed{\frac{2(q+3)}{(1+q)^2}}. \end{aligned}$$

The field is C^1 on all of $\mathbb{R}^3$, so its zero curl implies conservativity here. A direct verification is

$$\boxed{\mathbf{F} = \nabla\big(xyz + \log(1 + x^2 + y^2 + z^2)\big)}.$$

Check. Differentiating the displayed potential recovers all three components. At the origin the divergence is 6, agreeing with the unsimplified derivative sum.

Alternative. Recognize the potential first, then calculate the divergence by summing its three second partial derivatives. The requested curl and divergence still need to be supplied.

2. Green's theorem

Key observation. The exponential terms cancel in the scalar curl. Completing the square locates the disk; polar coordinates centered at $(1, 0)$ make the remaining odd term integrate to zero.

For $P = ye^{xy}$ and $Q = xe^{xy} + x^3/3 + xy^2$,

$$Q_x = e^{xy} + xye^{xy} + x^2 + y^2, \qquad P_y = e^{xy} + xye^{xy},$$

so $Q_x - P_y = x^2 + y^2$. Since C is positively oriented and the field is smooth on and inside C, Green's theorem gives

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_R (x^2 + y^2)\,\mathrm{d}A.$$

The region is $(x-1)^2 + y^2 \le 1$, $y \ge 0$. Set

$$x = 1 + \rho\cos\theta, \qquad y = \rho\sin\theta, \qquad \begin{cases} 0 \le \theta \le \pi, \\ 0 \le \rho \le 1. \end{cases}$$

The Jacobian is ρ, so

$$\begin{aligned} \iint_R (x^2 + y^2)\,\mathrm{d}A &= \int_0^{\pi}\int_0^{1} (1 + 2\rho\cos\theta + \rho^2)\rho\,\mathrm{d}\rho\,\mathrm{d}\theta \\ &= \int_0^{\pi} \left(\frac{1}{2} + \frac{2}{3}\cos\theta + \frac{1}{4}\right)\mathrm{d}\theta = \boxed{\frac{3\pi}{4}}. \end{aligned}$$

Check. Ordinary polar coordinates instead give $0 \le \theta \le \pi/2$, $0 \le r \le 2\cos\theta$, and

$$\int_0^{\pi/2} 4\cos^4\theta \, \mathrm{d}\theta = \frac{3\pi}{4}.$$

Alternative. Cartesian bounds $0 \le x \le 2$, $0 \le y \le \sqrt{2x - x^2}$ are correct. Any correct evaluation of the resulting Green double integral is acceptable.

3. Surface parametrizations

(a) The ellipsoid parametrization is

$$\boxed{\mathbf{r}(\theta, \phi) = (3\sin\phi\cos\theta,\ 2\sin\phi\sin\theta,\ \cos\phi)}, \qquad \begin{cases} 0 \le \theta \le 2\pi, \\ 0 \le \phi \le \pi/6. \end{cases}$$

Indeed, $x^2/9 + y^2/4 = \sin^2\phi$. On the upper half, $0 \le \phi \le \pi/2$, the cylinder condition is $\sin\phi \le 1/2$, hence $\phi \le \pi/6$.

(b) Use $\mathbf{r}(r, \theta) = (r\cos\theta, r\sin\theta, r)$. The first octant initially gives $0 \le \theta \le \pi/2$. The plane condition is

$$r \ge r\sin\theta + 1 \quad \Longleftrightarrow \quad r \ge \frac{1}{1 - \sin\theta}.$$

This is compatible with $r \le 2$ exactly when $\sin\theta \le 1/2$. Thus a complete answer is

$$\boxed{\mathbf{r}(r, \theta) = (r\cos\theta, r\sin\theta, r)}, \qquad \begin{cases} 0 \le \theta \le \pi/6, \\ \dfrac{1}{1 - \sin\theta} \le r \le 2. \end{cases}$$

(c) Interpolate between the two corresponding endpoints:

$$\boxed{\mathbf{r}(s, t) = (1 - t)\mathbf{a}(s) + t\mathbf{b}(s) = (s(1 - t),\ st,\ s(1 - t) + 2t)},$$

with $0 \le s \le 1$, $0 \le t \le 1$. At $t = 0$ and $t = 1$ the two specified lines are recovered; fixing s and varying t gives their joining segment.

Check. In (a), substitution gives the ellipsoid equation. In (b), the lower bound is exactly the slanted plane and the upper bound is $z = 2$. In (c), recover $s = x + y$ and $t = (z - x)/2$, so the interior is not multiply covered. Standard seam/pole coincidences on parameter boundaries in (a) do not change the covered surface.

Alternative. Any equivalent two-parameter representation with the correct image and full domain is valid. For example, graph coordinates are possible in (a), and the order of the two parameters may be reversed.

4. Scalar surface integral: setup only

Key observation. The radial coordinate is $u > 0$, so the exponential's quotient becomes v. The mixed trigonometric terms in the area factor cancel after squaring, but the parameter region is still triangular.

Write $c = \cos v$, $s = \sin v$. Then

$$\mathbf{r}_u = (c, s, v), \qquad \mathbf{r}_v = (-us, uc, u),$$

and

$$\mathbf{r}_u \times \mathbf{r}_v = u(s - vc, -c - vs, 1).$$

Its magnitude is

$$\begin{aligned} \|\mathbf{r}_u \times \mathbf{r}_v\| &= u\sqrt{(s - vc)^2 + (c + vs)^2 + 1} \\ &= u\sqrt{2 + v^2}. \end{aligned}$$

Since $x^2 + y^2 = u^2$ and $z/\sqrt{x^2 + y^2} = uv/u = v$,

$$f(\mathbf{r}(u, v)) = u^2 e^v.$$

Therefore the requested setup is

$$\boxed{\iint_S f \, \mathrm{d}\sigma = \int_1^2 \int_0^{u-1} u^3 e^v \sqrt{2 + v^2} \, \mathrm{d}v \, \mathrm{d}u}.$$

No evaluation is requested.

Check. The cross product's third component is $u \geq 1$, so the map is regular. The radius u and angle $0 \leq v \leq 1$ identify each surface point uniquely. The displayed integrand is a nonnegative scalar.

Alternative. Reversing integration order gives $\int_0^1 \int_{v+1}^2 u^3 e^v \sqrt{2 + v^2} \, \mathrm{d}u \, \mathrm{d}v$. Reversing the cross-product order gives the same magnitude.

5. Direct computation of oriented flux

Key observation. The cylinder is sheared in the x direction as z changes. The exponential part of the field is parallel to the cylinder's ruling direction $(1, 0, 1)$, so it contributes no normal flux.

Parametrize by

$$\mathbf{r}(\theta, z) = (z + \cos\theta, \sin\theta, z), \qquad \begin{cases} 0 \leq \theta \leq 2\pi, \\ 0 \leq z \leq 2 + \sin\theta. \end{cases}$$

The height bound is positive for every θ. The tangent vectors are

$$\mathbf{r}_\theta = (-\sin\theta, \cos\theta, 0), \qquad \mathbf{r}_z = (1, 0, 1),$$

so

$$\mathbf{n} = \mathbf{r}_\theta \times \mathbf{r}_z = (\cos\theta, \sin\theta, -\cos\theta).$$

For $g(x, y, z) = (x - z)^2 + y^2$, this equals $\frac{1}{2}\nabla g$ on S, so it points toward the exterior $g > 1$. At $\theta = 0$, $z = 1$, for example, the point is $(2, 0, 1)$ and $\mathbf{n} = (1, 0, -1)$ points outward.

Put $c = \cos\theta$, $s = \sin\theta$ and $E = e^{2z+c+s}$. Substitution gives

$$\mathbf{F}(\mathbf{r}) = (zc + E, zs, E), \qquad \mathbf{F}(\mathbf{r}) \cdot \mathbf{n} = zc^2 + Ec + zs^2 - Ec = z.$$

Thus

$$\begin{aligned}\iint_S \mathbf{F}\cdot\mathbf{N}\,\mathrm{d}\sigma &= \int_0^{2\pi}\int_0^{2+\sin\theta} z\,\mathrm{d}z\,\mathrm{d}\theta \\ &= \frac{1}{2}\int_0^{2\pi}(2+\sin\theta)^2\,\mathrm{d}\theta \\ &= \frac{1}{2}(8\pi+0+\pi) = \boxed{\frac{9\pi}{2}}.\end{aligned}$$

Check. The exponential vector is $E(1,0,1)$, whose dot product with $\mathbf{n}$ is zero. The remaining field has dot product z, agreeing with the explicit substitution. All z values are nonnegative, so the positive final sign is also consistent.

Alternative. A second parameter $v \in [0,1]$ may encode height through $z = v(2+\sin\theta)$. The corresponding cross product and Jacobian must then be included. It produces the same flux by direct parametrization.

6. Stokes' theorem

Key observation. The diagonal edge of the base triangle lifts to a parabolic arc, not a straight segment in the plane $z = 0$. On that arc the difficult terms form a one-variable total derivative whose endpoint values agree.

The downward orientation induces a clockwise boundary when projected onto the xy-plane. A single piecewise parametrization following $(0,0,0) \to (0,1,0) \to (1,0,0) \to (0,0,0)$ is

$$\mathbf{c}(t) = \begin{cases} (0,t,0), & 0 \le t \le 1, \\ (s, 1-s, s(1-s)), \quad s = t-1, & 1 \le t \le 2, \\ (3-t,0,0), & 2 \le t \le 3. \end{cases}$$

Each piece is on S, and the second piece is the lift of $x + y = 1$. Its derivative, piece by piece, is

$$\mathbf{c}\,'(t) = \begin{cases} (0,1,0), & 0 < t < 1, \\ (1,-1,1-2s), & 1 < t < 2, \\ (-1,0,0), & 2 < t < 3. \end{cases}$$

On the first and last pieces, respectively,

$$\mathbf{F}(\mathbf{c}(t)) = (t,0,0), \qquad \mathbf{F}(\mathbf{c}(t)) = (0,3-t,0),$$

so both dot products are zero. On the middle piece put $w = s(1-s)$. Then

$$\mathbf{F}(\mathbf{c}(t)) = \big((1-s)(e^w+w^2),\ s(e^w+w^2)+w,\ 2w^2\big),$$

and

$$\begin{aligned}\mathbf{F}(\mathbf{c}(t))\cdot\mathbf{c}\,'(t) &= (1-2s)(e^w+3w^2) - w \\ &= \frac{\mathrm{d}}{\mathrm{d}s}(e^w+w^3) - w.\end{aligned}$$

Stokes' theorem now gives the required boundary line integral:

$$\begin{aligned}\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N}\, \mathrm{d}\sigma &= \oint_{\partial S} \mathbf{F} \cdot \mathrm{d}\mathbf{r} \\ &= \int_0^1 \left[\frac{\mathrm{d}}{\mathrm{d}s}(e^{s(1-s)} + s^3(1-s)^3) - s(1-s) \right] \mathrm{d}s \\ &= \left[e^w + w^3\right]_{s=0}^{s=1} - \left[\frac{s^2}{2} - \frac{s^3}{3} \right]_0^1 = \boxed{-\frac{1}{6}}.\end{aligned}$$

Check. For an independent check only, $\mathbf{F} = \nabla(e^{xy} + xyz^2) + (0, z, 0)$, so $\nabla \times \mathbf{F} = (-1, 0, 0)$. The downward area vector of the graph is $(y, x, -1)\, \mathrm{d}x\, \mathrm{d}y$. Its dot product with this curl is $-y$, whose integral over the base triangle is $-1/6$.

Alternative. After identifying and parametrizing the boundary, recognize the displayed potential decomposition. Its gradient contribution has zero closed-curve integral, leaving $\oint z\, \mathrm{d}y$, which is $-\int_0^1 s(1-s)\, \mathrm{d}s$.

7. Conceptual questions

(a) The answers are

Expression	Type	Must be zero?
(i)	scalar	yes
(ii)	undefined	not applicable
(iii)	scalar	no

The field $f\mathbf{V}$ is a vector field, and the divergence of its curl is zero. In (ii), $\nabla \cdot \mathbf{V}$ is scalar, so taking its curl is not a defined operation in the course. For (iii), $f\nabla f$ is vector-valued; its divergence need not vanish. The choice $f = x$ gives $\nabla \cdot (f\nabla f) = \nabla \cdot (x, 0, 0) = 1$.

(b) The correct selections are **A and B**. Direct parametrization applies to the given smooth curve. Also $\mathbf{H} = \nabla(xyz)$, so the Fundamental Theorem of Line Integrals applies. The curve is open: its endpoints are $(0, 0, 0)$ and $(1, 1, 1)$. Neither Green's theorem nor Stokes' theorem applies directly to it as stated. No integral evaluation was requested.

(c) The local rotational axis is the positive x direction, with **counterclockwise** tendency viewed from the positive x-axis. Zero divergence indicates **neither** local net volume expansion nor local net volume compression. It does not force the curl to vanish. Likewise, a zero velocity at one point does not force its spatial derivatives to vanish.

Check. The local data in (c) are consistent, for example, with $\mathbf{v} = (0, -z, y)$ at the origin: its curl is $(2, 0, 0)$ and its divergence is zero.

Alternative. Equivalent physical language such as local contraction instead of sink is valid. Part (c) does not ask for angular speed.

Version D — Complete solutions

Instructor material. Key observations and checks are included for every problem. Sources and scope notes follow the solutions.

1. Curl, divergence, and global conservativity

Key observation. The domain omits the whole line $x = z$, $y = 0$. Zero curl on this domain does not, by itself, settle global conservativity.

Put $a = x - z$ and write

$$\mathbf{F}_\lambda = \nabla(xy + z^2) + \lambda \mathbf{W}, \qquad \mathbf{W} = \left(-\frac{y}{q}, \frac{a}{q}, \frac{y}{q}\right), \qquad q = a^2 + 2y^2.$$

Here $q_x = 2a$, $q_y = 4y$, and $q_z = -2a$. The mixed-component derivatives satisfy

$$(W_3)_y = (W_2)_z = \frac{a^2 - 2y^2}{q^2},$$
$$(W_1)_z = (W_3)_x = -\frac{2ay}{q^2},$$
$$(W_2)_x = (W_1)_y = \frac{2y^2 - a^2}{q^2}.$$

Thus $\nabla \times \mathbf{W} = \mathbf{0}$, and

$$\nabla \cdot \mathbf{W} = \frac{2ay - 4ay + 2ay}{q^2} = 0.$$

Since the base gradient has divergence 2,

$$\boxed{\nabla \times \mathbf{F}_\lambda = \mathbf{0}, \qquad \nabla \cdot \mathbf{F}_\lambda = 2 \quad \text{for every } \lambda.}$$

To decide conservativity, use the closed ellipse

$$\mathbf{c}(t) = \left(\cos t, \frac{\sin t}{\sqrt{2}}, 0\right), \quad 0 \le t \le 2\pi, \qquad \mathbf{c}\,'(t) = \left(-\sin t, \frac{\cos t}{\sqrt{2}}, 0\right).$$

It lies entirely in U because $q(\mathbf{c}(t)) = 1$. Direct substitution gives

$$\mathbf{W}(\mathbf{c}(t)) \cdot \mathbf{c}\,'(t) = \frac{1}{\sqrt{2}}, \qquad \oint_c \nabla(xy + z^2) \cdot \mathrm{d}\mathbf{r} = 0.$$

Consequently

$$\oint_c \mathbf{F}_\lambda \cdot \mathrm{d}\mathbf{r} = \lambda\sqrt{2}\pi.$$

A conservative field has zero integral around every closed curve. Hence $\lambda \neq 0$ is impossible. For $\lambda = 0$, the displayed base potential works on all of U. The complete answer is

$$\boxed{\mathbf{F}_\lambda \text{ is conservative on } U \iff \lambda = 0, \qquad f = xy + z^2 + C.}$$

Check. The same obstruction is visible in orthonormal coordinates $X = (x - z)/\sqrt{2}$, $Y = y$, $Z = (x + z)/\sqrt{2}$:

$$\mathbf{W} \cdot \mathrm{d}\mathbf{r} = \frac{1}{\sqrt{2}} \frac{-Y\ \mathrm{d}X + X\ \mathrm{d}Y}{X^2 + Y^2}.$$

A once-around circle in the XY-plane therefore has circulation $\sqrt{2}\pi$, agreeing with the explicit ellipse calculation. This coordinate observation is a check, not a needed topological theorem.

2. Green's theorem

Key observation. Green's theorem removes the exponential. The resulting single-variable integral is best paired with itself after a substitution.

With $P = ye^{xy} - y/(1+x^2)$ and $Q = xe^{xy}$,

$$Q_x = e^{xy} + xye^{xy}, \qquad P_y = e^{xy} + xye^{xy} - \frac{1}{1+x^2}.$$

The field is smooth near the whole region, and the prescribed boundary orientation is positive. Therefore

$$\begin{aligned} I &= \oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \int_0^1 \int_0^{\ln(1+x)} \frac{1}{1+x^2}\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^1 \frac{\ln(1+x)}{1+x^2}\,\mathrm{d}x. \end{aligned}$$

Use $x = (1-t)/(1+t)$, which exchanges 0 and 1. Then

$$\frac{\mathrm{d}x}{1+x^2} = -\frac{\mathrm{d}t}{1+t^2}, \qquad \ln(1+x) = \ln 2 - \ln(1+t).$$

Thus

$$I = \int_0^1 \frac{\ln 2 - \ln(1+t)}{1+t^2}\,\mathrm{d}t = \frac{\pi}{4}\ln 2 - I,$$

and

$$\boxed{I = \frac{\pi \ln 2}{8}.}$$

Check. Independently set $x = \tan\theta$, $0 \le \theta \le \pi/4$. If $h(\theta) = \ln(1+\tan\theta)$, the tangent subtraction formula gives $h(\theta) + h(\pi/4 - \theta) = \ln 2$. Integrating the pair gives the same answer. Numerically the value is approximately 0.2721982613.

Alternative. The reflection in the check is an equally valid evaluation of the double integral after Green's theorem has been used. Direct boundary integration alone does not meet the requested method.

3. Three surface parametrizations

Key observation. Use orthogonal projection for the tilted axis, straight rulings for the hyperboloid, and a dependent angular bound for the sphere.

(a) Let

$$\mathbf{e} = \frac{(1,1,0)}{\sqrt{2}}, \qquad \mathbf{b} = \frac{(1,-1,0)}{\sqrt{2}}, \qquad \mathbf{k} = (0,0,1).$$

The axis is parallel to $\mathbf{e}$, while $\mathbf{b}$ and $\mathbf{k}$ form an orthonormal basis of its perpendicular plane. The projection of $\mathbf{c}(t)$ onto the axis and its distance from the axis are

$$\mathbf{p}(t) = (\mathbf{c}(t)\cdot\mathbf{e})\mathbf{e} = \left(\frac{t}{2}, \frac{t}{2}, 0\right), \qquad R(t) = \sqrt{\frac{t^2}{2} + t^4}.$$

Rotating a point about the axis sweeps the entire circle with this center and radius. Hence one answer is

$$\boxed{\mathbf{r}(t,\theta) = \mathbf{p}(t) + R(t)\big(\mathbf{b}\cos\theta + \mathbf{k}\sin\theta\big)}, \qquad \begin{cases} 1 \le t \le 2, \\ 0 \le \theta \le 2\pi. \end{cases}$$

Its angular starting point need not be the original generating point; each whole circle is unchanged by a shift of angular origin. For a parametrization with $\mathbf{r}(t,0)=\mathbf{c}(t)$ instead, one can use

$$\mathbf{r}(t,\theta)=\mathbf{p}(t)+\left(\frac{t}{\sqrt{2}}\cos\theta+t^2\sin\theta\right)\mathbf{b}+\left(t^2\cos\theta-\frac{t}{\sqrt{2}}\sin\theta\right)\mathbf{k}.$$

This rotates the perpendicular vector $(t/\sqrt{2})\mathbf{b}+t^2\mathbf{k}$ with unchanged length.

(b) Start at $(\cos u,\sin u,0)$ and take a straight line with direction $(-\sin u,\cos u,1)$. This yields

$$\boxed{\mathbf{r}(u,v)=(\cos u-v\sin u,\ \sin u+v\cos u,\ v)},\qquad \begin{cases}0\le u\le 2\pi,\\ 0\le v\le 1.\end{cases}$$

At fixed u this is affine in v, as required. Moreover,

$$x^2+y^2=(\cos u-v\sin u)^2+(\sin u+v\cos u)^2=1+v^2,$$

so it lies on the hyperboloid. At each height $z=v$, the first two coordinates are a rotation and dilation of the unit circle, and thus cover the complete circle $x^2+y^2=1+v^2$. This proves full coverage.

(c) Use the usual sphere parametrization

$$\boxed{\mathbf{r}(\phi,\theta)=(2\sin\phi\cos\theta,2\sin\phi\sin\theta,2\cos\phi)}.$$

The extra inequality forces $z\ge 0$, so $0\le\phi\le\pi/2$. It becomes

$$\cos\phi\ge\sin\phi\big(|\cos\theta|+|\sin\theta|\big).$$

The complete bounds are therefore

$$\boxed{\begin{cases}0\le\theta\le 2\pi,\\ 0\le\phi\le\arctan\left(\dfrac{1}{|\cos\theta|+|\sin\theta|}\right).\end{cases}}$$

The denominator is at least 1, so this expression is defined at every azimuth and correctly includes all four quadrants.

Check. For (a), $\mathbf{r}\cdot\mathbf{e}=t/\sqrt{2}$ and $\|\mathbf{r}-\mathbf{p}(t)\|=R(t)$, precisely the invariants of each rotation orbit; the axis coordinate is strictly increasing with t. For (b), $\mathbf{r}_u\times\mathbf{r}_v=(\cos u-v\sin u,\sin u+v\cos u,-v)$ has squared norm $1+2v^2>0$. For (c), substitution gives sphere radius 2, and the inequality was transformed equivalently, so the bounds neither omit nor add a patch. The usual pole and angular-seam repetitions are harmless.

Alternative. In (b), the other family of straight rulings, obtained by reversing the tangential horizontal direction, is also valid. A generic height-and-angle parametrization without straight parameter curves does not meet that part's additional condition. Equivalent two-parameter maps and equivalent complete bounds are otherwise valid.

4. A scalar surface integral: setup only

Key observation. The map doubles the polar angle. Its full parameter annulus covers the same conical surface twice, and $\sqrt{u^2}$ is $|u|$.

The required derivatives are

$$\mathbf{r}_u = (2u, 2v, 2u), \qquad \mathbf{r}_v = (-2v, 2u, 2v),$$

$$\mathbf{r}_u \times \mathbf{r}_v = (4v^2 - 4u^2, -8uv, 4u^2 + 4v^2), \qquad \|\mathbf{r}_u \times \mathbf{r}_v\| = 4\sqrt{2}(u^2 + v^2).$$

The substituted function is

$$f(\mathbf{r}(u, v)) = \sqrt{2u^2} + e^{u^2+v^2} = \sqrt{2}|u| + e^{u^2+v^2}.$$

To examine coverage, put $u = s\cos\alpha$, $v = s\sin\alpha$. Then

$$\mathbf{r} = (s^2\cos 2\alpha, s^2 \sin 2\alpha, s^2), \qquad 1 \le s \le 2.$$

The full range $0 \le \alpha \le 2\pi$ travels around each surface circle twice. The half-annulus $v \ge 0$, equivalently $0 \le \alpha \le \pi$, covers the geometric surface once except on its seam. Its image is $z = \sqrt{x^2 + y^2}$, $1 \le z \le 4$.

The polar Jacobian in the *parameter plane* is s, in addition to the already computed surface factor. Thus a complete unevaluated setup is

$$\boxed{\iint_S f \, \mathrm{d}\sigma = \int_0^{\pi} \int_1^2 \left(\sqrt{2}s|\cos\alpha| + e^{s^2}\right) 4\sqrt{2}s^3 \, \mathrm{d}s \, \mathrm{d}\alpha.}$$

Check. The cross-product norm never vanishes on the annulus. Equality of two images forces the same s and angles differing by a multiple of π, so there are exactly two preimages in the full annulus away from seams. As a separate check, the once-around cone map $(w\cos\beta, w\sin\beta, w)$ has area factor $\sqrt{2}w$; $w = s^2$, $\beta = 2\alpha$ reproduce $4\sqrt{2}s^3 \, \mathrm{d}s \, \mathrm{d}\alpha$.

Alternative. One-half of the full-annulus parameter integral is equivalent:

$$\frac{1}{2}\int_0^{2\pi} \int_1^2 \left(\sqrt{2}s|\cos\alpha| + e^{s^2}\right) 4\sqrt{2}s^3 \, \mathrm{d}s \, \mathrm{d}\alpha.$$

This is a correction for repeated coverage, not for orientation. A scalar surface integral uses the cross-product magnitude.

5. Direct flux

Key observation. The field is radial, but the upward normal changes from pointing partly with the field to pointing partly against it. Reciprocal radii pair these contributions exactly.

Use the direct parametrization

$$\mathbf{r}(r, \theta) = (r\cos\theta, r\sin\theta, 1 + r^2), \qquad \begin{cases} \frac{1}{2} \le r \le 2, \\ 0 \le \theta \le 2\pi. \end{cases}$$

Then

$$\mathbf{r}_r = (\cos\theta, \sin\theta, 2r), \quad \mathbf{r}_\theta = (-r\sin\theta, r\cos\theta, 0),$$

$$\mathbf{n} = \mathbf{r}_r \times \mathbf{r}_\theta = (-2r^2\cos\theta, -2r^2\sin\theta, r).$$

Its third component r is positive everywhere on the parameter domain, so it gives the required upward orientation. With $D(r) = r^4 + 3r^2 + 1$,

$$\mathbf{H}(\mathbf{r}) = \frac{(r\cos\theta, r\sin\theta, 1+r^2)}{D(r)^{3/2}}, \qquad \mathbf{H}(\mathbf{r})\cdot\mathbf{n} = \frac{r(1-r^2)}{D(r)^{3/2}}.$$

Therefore the flux is

$$\Phi = \int_0^{2\pi}\int_{1/2}^{2} \frac{r(1-r^2)}{(r^4+3r^2+1)^{3/2}}\ \mathrm{d}r\ \mathrm{d}\theta.$$

One way to evaluate it is to observe that

$$\frac{\mathrm{d}}{\mathrm{d}r}\frac{1+r^2}{\sqrt{r^4+3r^2+1}} = \frac{r(r^2-1)}{(r^4+3r^2+1)^{3/2}}.$$

The endpoint values of this primitive are both $5/\sqrt{29}$:

$$\boxed{\Phi = -2\pi\left[\frac{1+r^2}{\sqrt{r^4+3r^2+1}}\right]_{1/2}^{2} = 0.}$$

Check. There is an independent evaluation without guessing a primitive. Put $g(r) = r(1-r^2)/(r^4+3r^2+1)^{3/2}$. Direct algebra gives

$$\frac{g(1/s)}{s^2} = -g(s), \qquad \int_1^2 g(r)\ \mathrm{d}r = -\int_{1/2}^{1} g(s)\ \mathrm{d}s.$$

The two radial integrals cancel. The signed density is positive for $r < 1$ and negative for $r > 1$, so zero is compatible with the field being nonzero everywhere. Numerical integration also gives zero to numerical precision. No divergence theorem or added surface is used.

6. Stokes' theorem

Key observation. The sphere and cylinder intersect in a nonplanar curve. A half-angle identity makes its height elementary and nonnegative.

The cylindrical condition is $x^2 + y^2 \le 2x$. Its boundary on the sphere has

$$x^2 + y^2 = 2x, \qquad z^2 = 4 - 2x, \qquad z \ge 0.$$

Parametrize the projected circle by $x = 1 + \cos t$, $y = \sin t$. Then

$$z = \sqrt{2 - 2\cos t} = 2\sin(t/2), \qquad 0 \le t \le 2\pi,$$

where the sine is nonnegative on this interval. Hence the complete boundary is

$$\mathbf{c}(t) = (1 + \cos t, \sin t, 2\sin(t/2)), \qquad 0 \le t \le 2\pi.$$

There is no additional equatorial arc: on $z = 0$, the sphere has $x^2 + y^2 = 4$, while the cylinder inequality demands $4 \le 2x$; the only possible point is $(2, 0, 0)$. This is already the starting and ending point of $\mathbf{c}$.

On the rest of S the outward sphere normal has positive third component. Thus the counterclockwise projected circle gives the induced boundary orientation. The corner of the closed boundary at $(2,0,0)$ is allowed for a piecewise smooth boundary. Its derivative is

$$\mathbf{c}\,'(t) = (-\sin t, \cos t, \cos(t/2)).$$

Write $s = \sin(t/2)$, $c = \cos(t/2)$. On this curve,

$$\mathbf{F}(\mathbf{c}(t)) = (0, 4s^2, -4s^2 c),$$

and

$$\mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}\,'(t) = 4s^2(c^2 - s^2) - 4s^2c^2 = -4s^4.$$

Stokes' theorem therefore gives

$$\begin{aligned}\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N}\, d\sigma = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} &= -4\int_0^{2\pi} \sin^4(t/2)\, dt \\ &= -8\int_0^{\pi} \sin^4 u\, du = \boxed{-3\pi}.\end{aligned}$$

For the last integral, $\sin^4 u = (3 - 4\cos 2u + \cos 4u)/8$ gives $\int_0^{\pi} \sin^4 u\, du = 3\pi/8$.

Check. Independently, $\nabla \times \mathbf{F} = (-3z, 0, 0)$. On the upper sphere graph, the outward area vector is $(x/z, y/z, 1)\, dx\, dy$ where $z > 0$. The dot product is $-3x$, which extends continuously to the lone equatorial boundary point. Over the disk centered at $(1,0)$ of radius 1, its integral is -3π. This check agrees with Stokes; the required solution above actually uses the boundary line integral.

7. Operators, physical meaning, and a global obstruction

Key observation. Multiplying a curl by a variable scalar can create divergence. Also, a global vector potential requires more than local zero divergence on a domain with an excluded point.

(a) The expression is a defined **scalar field**. The divergence product rule and the identity $\nabla \cdot (\nabla \times \mathbf{G}) = 0$ give

$$\boxed{\nabla \cdot (f\, \nabla \times \mathbf{G}) = \nabla f \cdot (\nabla \times \mathbf{G}).}$$

It need not vanish. For example,

$$f = z, \qquad \mathbf{G} = (-y/2, x/2, 0), \qquad \nabla \times \mathbf{G} = (0,0,1),$$

so $\nabla \cdot (f\, \nabla \times \mathbf{G}) = \nabla \cdot (0,0,z) = 1$.

(b) Put $\rho = \sqrt{x^2 + y^2 + z^2} > 0$. Direct differentiation gives

$$\nabla \cdot \mathbf{H} = \left(\rho^{-3} - 3x^2\rho^{-5}\right) + \left(\rho^{-3} - 3y^2\rho^{-5}\right) + \left(\rho^{-3} - 3z^2\rho^{-5}\right) = 0.$$

Also $\mathbf{H} = \nabla(-1/\rho)$, so $\nabla \times \mathbf{H} = \mathbf{0}$ on U_0. Thus at every point of U_0 these operators predict **neither local volume expansion nor compression, and no local rotational tendency**. They do not say that the velocity is zero or that the motion cannot stretch in some directions while compressing in others.

For a sphere of radius R about the origin, however,

$$\mathbf{H} \cdot \mathbf{N} = \frac{1}{R^2}, \qquad \iint_{\rho = R} \mathbf{H} \cdot \mathbf{N}\, d\sigma = 4\pi.$$

This outward flux encloses the excluded origin. No conclusion of zero flux through such a sphere follows just from divergence zero at every point where the field is defined. There is no contradiction with the local source-free interpretation away from the origin.

(c) **No such global A exists.** Suppose it did. Split the unit sphere into its upper and lower hemispheres S_+ and S_-, both oriented outward. Their common equator is counterclockwise when induced by S_+ and clockwise when induced by S_-. Since $\mathbf{A}$ would be smooth near both hemispheres, Stokes' theorem would imply

$$\begin{aligned}\iint_{S^2} \mathbf{H}\cdot\mathbf{N}\,\mathrm{d}\sigma &= \iint_{S_+} \nabla\times\mathbf{A}\cdot\mathbf{N}\,\mathrm{d}\sigma + \iint_{S_-} \nabla\times\mathbf{A}\cdot\mathbf{N}\,\mathrm{d}\sigma \\ &= \oint_{C} \mathbf{A}\cdot\mathrm{d}\mathbf{r} + \oint_{-C} \mathbf{A}\cdot\mathrm{d}\mathbf{r} = 0.\end{aligned}$$

But the actual flux, computed in (b), is 4π. This contradiction rules out a C^2 vector potential defined on all of U_0. The argument uses only Stokes on two ordinary smooth hemispheres and cancellation of their common oriented boundary.

Check. For the unit sphere the flux calculation can be checked directly with $\mathbf{r}(\phi,\theta) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi)$: the outward cross product $\mathbf{r}_\phi\times\mathbf{r}_\theta = \sin\phi\,\mathbf{r}$ gives $\int_0^{2\pi}\int_0^{\pi}\sin\phi\;\mathrm{d}\phi\;\mathrm{d}\theta = 4\pi$. This proves the necessary obstruction without a theorem about domains or a divergence-theorem argument.

Version D — Instructor notes

Not part of the student-facing questions.

Purpose and alignment. This is an untimed challenge set, deliberately above the expected actual Exam 3 level. It preserves the outline's seven sections and nominal weights $10+12+12+12+20+20+14=100$, while following the revised compact A–C practice format without displayed point counts or answer boxes. Problem 3 has three two-parameter maps; Problem 4 supplies its map and asks only for a setup; Problem 5 requires a direct oriented-flux calculation; Problem 6 requires Stokes and an actual one-parameter boundary integral. Green's theorem is required in Problem 2. For Problem 7, a possible division is $4+4+6$. No timing claim is made.

What exceeds normal expectations.

1. The zero-curl test must be distinguished from global conservativity on a domain missing a tilted line. Students must construct an obstructing closed curve. No fundamental-group or topology theorem is assumed.

2. The final integral requires discovery of a fractional-linear substitution, or its trigonometric-reflection equivalent. This is a contest-level integration observation after an ordinary Green setup.

3. Rotation about an oblique axis requires constructing circle centers and radii. The hyperboloid must be recognized as ruled, not merely parametrized in the most familiar way. The sphere has a genuinely azimuth-dependent angular bound. Each uses elementary vectors and algebra.

4. The supplied regular map is not one-to-one on its full domain. Correcting multiplicity and preserving an absolute value exceed the routine surface-integral setups in the outline. The prompt explicitly requires counting the geometric image once.

5. The signed radial flux cancels under reciprocal radii. The insight is in the evaluation, not a large field or lengthy cross product. No divergence theorem or solid-angle theorem is assumed.

6. The boundary is a sphere–cylinder intersection with a corner, not a standard horizontal circle. Discovering the half-angle parametrization and ruling out an extra equatorial component are the extensions.

7. The product identity itself is familiar. The contrast between local zero divergence and a global obstruction to being a curl is beyond ordinary Exam 3 practice. Two applications of the already taught Stokes theorem supply the complete argument; no cohomology result is needed.

Origins and adaptations. The following notes distinguish specific mathematical seeds from original constructions. None of the student prompts is copied from a contest.

- **Problem 1: original construction.** The tilted-line field was constructed so its curl vanishes and its divergence is constant for every λ, while a closed ellipse detects its nonzero circulation. It uses the standard circulation obstruction also underlying the source listed for Problem 7.

- **Problem 2: Putnam 2005, A5.** `https://kskedlaya.org/putnam-archive/2005.pdf` The final one-variable logarithmic integral and the involutive substitution are the specific borrowed seed. The region, vector field, Green's-theorem formulation, and required intermediate setup are an original adaptation.

- **Problems 3–4: original constructions.** The chosen oblique revolution, ruled-hyperboloid requirement, sphere patch, and double-covered conical annulus were constructed for this version. The complex-square coordinate identity in Problem 4 is used only as elementary real algebra; students need no complex analysis.

- **Problem 5: strategy inspired by Putnam 1985, B5.** `https://kskedlaya.org/putnam-archive/1985.pdf` The borrowed strategy is reciprocal-variable pairing. The paraboloid, annular bounds, inverse-square field, and direct flux calculation are new. None of that source's Gaussian-integral machinery is used.

- **Problem 6: Viviani-curve geometry.** Oliver Knill, Harvard Math 136, Unit 1 homework, Problem 1.1: `https://people.math.harvard.edu/~knill/teaching/math136/handouts/homework01.pdf` The sphere–cylinder intersection is the geometric seed, rescaled and restricted to its upper loop. The field, flux question, boundary orientation task, and Stokes calculation are new. Curvature and torsion from that source are irrelevant here and are not assumed.

- **Problem 7(c): inspired by Putnam 1987, A5.** `https://kskedlaya.org/putnam-archive/1987.pdf` That problem uses a nonzero-circulation obstruction to a curl-free extension. Here the analogous obstruction is nonzero sphere flux, excluding a global vector potential. The proof uses two hemispheres and ordinary Stokes. Parts (a)–(b) and their linkage with Problem 5 are original.

Other research consulted, not directly adapted. The official IMC 2026 Problem 7 materials (`https://www.imc-math.org.uk/?item=prob7s&section=problems&year=2026`) were considered for overlap and ruled-region ideas; its inequality is not used. The official ETS *GRE Mathematics Practice Book*, form GR3768, including the symmetry integral in Question 63, was checked as a comparison for observation-based calculus: `https://www.ets.org/content/dam/ets-org/pdfs/gre/practice-book-math.pdf`. The official IMO 1961 set (`https://www.imo-official.org/assets/documents/problems/1961/1961_eng.pdf`) and Kyoto University's 2018 master's entrance basic-mathematics paper, especially its first improper-integral problem (`https://www.math.kyoto-u.ac.jp/files/master_exams/2017math_kiso.pdf`), were also considered. Their original questions were not inserted: the selected tasks keep the Exam 3 vector-calculus topics central.

Verification record. Each solution was recomputed independently by a second reader. Checks include the explicit circulation obstruction in Problem 1, two evaluations of Problem 2, algebra and coverage tests for all three maps in Problem 3, an independent cone parametrization for Problem 4, both a primitive and reciprocal pairing for Problem 5, a direct surface check of Problem 6, and direct sphere flux plus two-hemisphere Stokes for Problem 7. Numerical quadrature independently agrees with $\pi \ln 2/8$, 0, and -3π in Problems 2, 5, and 6. The mathematical content of Versions A–C and the original outline is unchanged.